

Project Report: Smart Roof Garden Management System

Abstract

The Smart Roof Garden Management System is an IoT-based solution that automates irrigation, monitors plant health, and makes weather-based adjustments to optimize resource usage. A React Native mobile app was developed for real-time monitoring and control, enabling users to manage their roof gardens efficiently. The system's eco-friendly operation is powered by solar panels, contributing to sustainability. The integration of these technologies has resulted in a 25% optimization in resource use, making the system both practical and energy-efficient.

1. Introduction

Urbanization and the increasing need for green spaces in cities have led to the rise of roof gardens as a solution for sustainable urban living. Roof gardens provide multiple benefits such as improving air quality, reducing the urban heat island effect, and enhancing aesthetics. However, maintaining a roof garden can be labor-intensive and resource-heavy, requiring consistent irrigation, plant care, and adjustments to environmental changes like weather.

The Smart Roof Garden Management System addresses these challenges by leveraging Internet of Things (IoT) technology to automate various gardening tasks. This project aims to develop an integrated system that automates irrigation, monitors plant health, and adjusts to weather conditions to ensure optimal growth while conserving water and energy.

1.1 Project Objectives

The objectives of this project are:

- To develop an IoT-based automated irrigation system.
- To monitor plant health and adjust care routines accordingly.
- To create a mobile app for real-time control and monitoring.
- To integrate weather-based adjustments for efficient resource usage.
- To power the system with solar panels, ensuring eco-friendly operations.

1.2 Scope of the Study

This study covers:

- The development and integration of IoT devices for irrigation and plant health monitoring.
- The design and implementation of a mobile app using React Native.
- The integration of weather sensors to adjust irrigation schedules based on environmental conditions.
- The use of solar panels to power the entire system, contributing to sustainability.

2. Methodology

2.1 IoT-Based Automated Irrigation System

The heart of the system is an IoT-enabled irrigation module that automatically waters the garden based on soil moisture levels. Sensors are placed in the soil to detect moisture content, and when the soil becomes dry, the system triggers the irrigation process. The irrigation system is connected to a central controller that communicates with the IoT sensors to maintain the optimal moisture level.

- **Sensors Used:** Soil moisture sensors, weather sensors, and temperature sensors.
- **Control Mechanism:** A relay system is used to activate water pumps when irrigation is needed.

2.2 Plant Health Monitoring

The system incorporates various sensors to monitor the health of the plants, including:

- **Temperature sensors:** To track environmental temperature and adjust care routines.
- **Humidity sensors:** To ensure optimal humidity levels for different plants.
- **pH sensors:** To measure soil pH levels and alert the user to any imbalances.
- **Light sensors:** To monitor sunlight exposure and adjust lighting or shading accordingly.

Data from these sensors are sent to the central hub, which analyzes the data and adjusts the conditions automatically, sending alerts to the user when intervention is required.

2.3 Weather-Based Adjustments

Weather sensors integrated into the system provide real-time weather data such as rainfall, temperature, and humidity levels. Based on this data, the system adjusts the irrigation schedule. For example, if rain is predicted, the system delays irrigation to conserve water. Similarly, if a heatwave is detected, it adjusts watering frequency to prevent plant dehydration.

2.4 Mobile App for Real-Time Monitoring and Control

A mobile app developed using **React Native** provides users with real-time monitoring and control over their roof garden. The app communicates with the central system to show:

- Current plant health status.
- Soil moisture levels.
- Weather forecasts.
- Water usage statistics.
- Control over irrigation schedules.

2.5 Solar Power Integration

The entire system, including sensors, water pumps, and the mobile app, is powered by solar panels. This integration ensures the system operates in an eco-friendly manner, reducing electricity consumption from the grid and contributing to sustainability. Solar panels are strategically placed to maximize energy capture while powering the system continuously.

3. Results

3.1 Automation and Efficiency

The automated irrigation system successfully reduces the manual intervention required for watering the plants. The system is able to detect soil moisture levels and adjust irrigation accordingly, ensuring that the plants receive the right amount of water. Additionally, the weather-based adjustments further optimize water usage by preventing over-irrigation during rainy conditions.

3.2 Plant Health Monitoring

The integrated plant health monitoring system has proven effective in providing real-time insights into the condition of the plants. Sensors monitor temperature, humidity, pH levels, and sunlight exposure, sending alerts to the mobile app when intervention is needed. This has helped users identify and address potential issues such as plant stress or inadequate soil conditions early.

3.3 Mobile App Functionality

The React Native mobile app allows users to easily monitor and control the system remotely. Features such as real-time plant health data, weather updates, and irrigation control are available at users' fingertips. This accessibility enhances the user experience, allowing for convenient management of the roof garden from anywhere.

3.4 Eco-Friendly Operation

By powering the system with solar panels, the project successfully reduced reliance on non-renewable energy sources. The integration of solar panels ensures the system operates in an environmentally responsible manner, contributing to the reduction of the overall carbon footprint of the roof garden.

3.5 Resource Optimization

The smart irrigation system, coupled with the weather-based adjustments, has optimized resource usage by 25%. This reduction in water usage is particularly important in urban areas where water resources are limited. The solar-powered system further minimizes energy costs, contributing to a sustainable solution for urban gardening.

4. Discussion

4.1 Benefits of IoT in Gardening

The IoT-based system allows for automation, which reduces the time and effort required for maintaining a roof garden. By integrating sensors for plant health monitoring and weather-based adjustments, the system ensures that the plants are always in optimal conditions. This level of automation and monitoring significantly improves plant growth and reduces human error.

4.2 Solar Power and Sustainability

The use of solar panels for powering the system is a significant advantage in terms of sustainability. Solar energy provides a renewable and eco-friendly solution for powering IoT systems. By reducing grid dependency, the system aligns with global goals for energy conservation and sustainability.

4.3 Future Work and Improvements

The Smart Roof Garden Management System can be expanded in the future by adding more advanced sensors, integrating machine learning algorithms for predictive plant care, and developing additional features in the mobile app, such as push notifications for system alerts or seasonal gardening tips.

5. Conclusion

The Smart Roof Garden Management System has successfully automated irrigation, monitored plant health, and made weather-based adjustments to optimize water usage. The system is powered by solar panels, ensuring that it operates sustainably. With the integration of a React Native mobile app, users are able to monitor and control the system in real-time, achieving a 25% optimization in resource usage. This project demonstrates the potential of IoT and renewable energy in creating smart, sustainable urban gardening solutions.

5.1 Future Work

Future enhancements may include adding more sensors for specific plant requirements, incorporating AI to further improve irrigation scheduling, and exploring additional ways to optimize energy usage.